

FARMANA DITYA ALYA SAFITRI

Applied Geographic Information Systems · Universitas Gadjah Mada

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Final-year Applied GIS student with hands-on experience across national and regional geospatial agencies, spanning cadastral surveying, spatial data standardisation, thematic mapping, and web GIS development. Final project applies Random Forest classification and spatial metrics to multi-temporal Landsat imagery. Teaching assistant for three applied GIS practicum courses.

EDUCATION

Universitas Gadjah Mada — Sleman, D.I. Yogyakarta Aug 2022 – Aug 2026 (expected)

Bachelor of Applied Science, Geographic Information Systems · GPA 3.42 / 4.00

- Coursework in remote sensing, spatial analysis, cartography, web & mobile GIS, and GIS applications for disaster management and transport logistics.

SMK Negeri 2 Surakarta — Surakarta Jul 2019 – Jul 2022

Geomatics and Geospatial Program

- PUPR–LPJK professional competency certification in geomatics surveying.

PROFESSIONAL EXPERIENCE

Universitas Gadjah Mada — Sleman, D.I. Yogyakarta Feb 2026 – Jun 2026

Teaching Assistant · Applied GIS Study Program

- Assisted 65+ students across three practicum courses: Field Work Practice, GIS for Disaster Management, and GIS for Transport & Logistics.
- Guided students through spatial analysis workflows in ArcGIS and Python, troubleshooting technical issues during lab sessions.

Badan Informasi Geospasial (BIG) — Bogor Sep 2025 – Dec 2025

Geospatial Internship · Direktorat Atlas dan Penggunaan Informasi Geospasial (DAPIG)

- Processed and standardised road and river geospatial datasets across multiple provinces to support development of the Peta NKRI 2025.
- Digitised development zones and strategic nodes for 7 regional planning areas, aligned with RPJPN 2025–2045.
- Developed thematic and connectivity maps through spatial analysis, multi-layer integration, and map layouting.

Dinas Pertanahan dan Tata Ruang — Yogyakarta Feb 2025 – Jun 2025

GIS Internship · Bidang Tata Ruang

- Evaluated zoning regulations for outdoor advertisements (Reklame) and assessed the Pepadhang Resik Jogja WebGIS for system improvement.
- Contributed to Loka Tata WebGIS development: spatial data collection, land-use conformity (TKKPR) digitisation, and database integration.
- Digitised 1,000+ building footprints and produced visualisations supporting the building numbering survey with PPIDS UGM.

Badan Pertanahan Nasional (BPN) — Surakarta Jan 2024 – Mar 2024

Internship

- Processed satellite imagery and elevation data for urban mapping using Google Earth Engine and ENVI.
- Verified 30+ land use requests (PTP/KKPR), including on-site inspections.
- Improved 500+ parcel shapefiles with topology corrections and attribute completion.

Badan Pertanahan Nasional (BPN) — Boyolali Sep 2021 – Jan 2022

Student Internship

- Conducted cadastral surveying and mapping for 20+ land parcels using geodetic GNSS, and managed land measurement documents in the BPN web system.

PROJECTS

Final Project — Urban Sprawl Mapping Using the Shannon Entropy Index — Sukoharjo Regency 2025 – 2026

- Classified built-up land from multi-temporal Landsat imagery (2011, 2018, 2025) using Random Forest on Google Earth Engine, achieving Kappa coefficients of 0.81–0.85.
- Measured sprawl with a two-level Shannon Entropy index and mapped all 12 districts into a development typology combining entropy with population density.
- Village Spatial Planning Fieldwork, Ngargosari, Kulon Progo** (2024) — Field data acquisition, land-use mapping, and population and building surveys for GIS-based village planning.
- Bookish Brew WebGIS** (2024) — Laravel-based WebGIS mapping bookstores and cafés in Yogyakarta, built with PostGIS and Leaflet.
- SPBU Finder Mobile App** (2024) — React Native app displaying gas station locations, featuring interactive maps, marker clustering, and data entry.

ORGANISATIONAL EXPERIENCE

- STARGIS — Student Association of Remote Sensing and GIS** (Mar 2023 – Mar 2025) — Social & Community Division Staff; organised fundraising and donation distribution for an orphanage.
- GEOSTAV** (Jul 2023 – Sep 2023) — Facilitator; led orientation for 20 incoming GIS students and prepared learning materials.

SKILLS

GIS & Remote Sensing ArcGIS, QGIS, Google Earth Engine, ENVI

Surveying Geodetic GNSS, Total Station, Theodolite, Leveling

Programming & Web Python, Laravel, PostGIS, Leaflet, React Native

Other Microsoft Office · Indonesian (native), English (professional working)